

Where the Shortcut Sits — Full Project Report

Location-dependent failure of ROI masking for in-ROI artifacts, and what to do instead

Project repository: https://github.com/hossam1999/Where_shortcut_sits

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1 Executive summary

In plain words. Medical AI models often “cheat” by looking at things that are not the disease—hair on the skin, measurement marks drawn by the sonographer, food debris in the gut. The usual fix is to show the model only the lesion (“masking”). We proved, in four diseases, that this fix fails when the cheating cue sits *on the lesion itself*: masking removes everything except the cue. We explained why with a simple formula, and we built and tested remedies. The strongest practical remedy combines masking, a new “erase” step that needs no annotation of the artifact, and re-balancing of the training data; which remedy works best depends on what actually carries the cheating cue.

1. **Succeeded The location effect.** Masking helps far more for artifacts outside the region of interest (ROI) than inside, in dermoscopy (hair), thyroid and ovarian ultrasound (calipers) and capsule endoscopy (debris): crossover +0.15 to +0.37 reversed-test AUROC, all significant after Holm correction; replicated on four frozen backbones, with fine-tuned ResNet-50 and ViT, and robust to the artifact-label definition.
2. **Succeeded Masking can harm.** When masking also removes disease context it hurts: real dermoscopy hair, controlled sweeps (ovary −0.20, capsule −0.10, dermoscopy rulers), capsule fine-tuning (−0.28), and on the *unaltered* thyroid test set (−0.10 on shortcut-conflicting cases).
3. **Succeeded Theory.** One closed-form expression explains all these signs (agreement with simulation 0.009 AUROC). Fitted only to each model’s clean and correlated results, it matches the held-out hard-test results of real models (average error 0.039 AUROC) and the location gap in 14 dataset/model pairs (correlation 0.997; 13 of 14 signs).
4. **Mixed The remedy (U-MtE).** Works where the artifact is a distinct overlay (thyroid +0.13 to +0.22 over masking on three backbones; +0.14 end-to-end with ResNet-50); erasure beats augmentation (3/4 cohorts); fails when the artifact resembles the disease (capsule) unless disease-protected, and cannot help when the shortcut lives in correlates of the artifact (hair, chest drains) — there label-based reweighting wins.
5. **Succeeded Foundation models.** The location effect is present in zero-shot vision–language models before any training; describing the artifact in text and erasing it does not work, our image-side method does.
6. **Mixed One procedure for all cases.** A validation-driven selector beats masking everywhere and matches the best candidate in each cohort, but is not uniformly better than DFR.
7. **Succeeded Reproducibility.** Pre-registration before every experiment; the original pilot replicated (73/88 numbers exactly); every table and figure generated by scripts; a two-command public tool.

2 The question

In plain words. If an AI model learns to recognise a disease from a mark that happens to accompany it, removing everything except the lesion should stop the cheating—unless the mark is on the lesion. We measure exactly when and how much.

A *trap* trains a classifier where the artifact accompanies one class 90% of the time and tests on a reversed environment where it accompanies the other class 90% of the time. A model that relies on the artifact collapses on the reversed test. We compare mitigations by reversed-test AUROC, and always also report correlated and clean AUROC to detect “gaming” (a method that merely flips the shortcut). The overlap $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$ defines in-ROI (Trap A, $r \geq 0.5$) and out-of-ROI (Trap B, $r < 0.1$) artifacts.

3 Data and artifacts

In plain words. We used five public medical imaging collections. Below are real examples: the green outline is the region of interest (lesion, nodule, tumour, lungs) and red marks the artifact the model could cheat with.

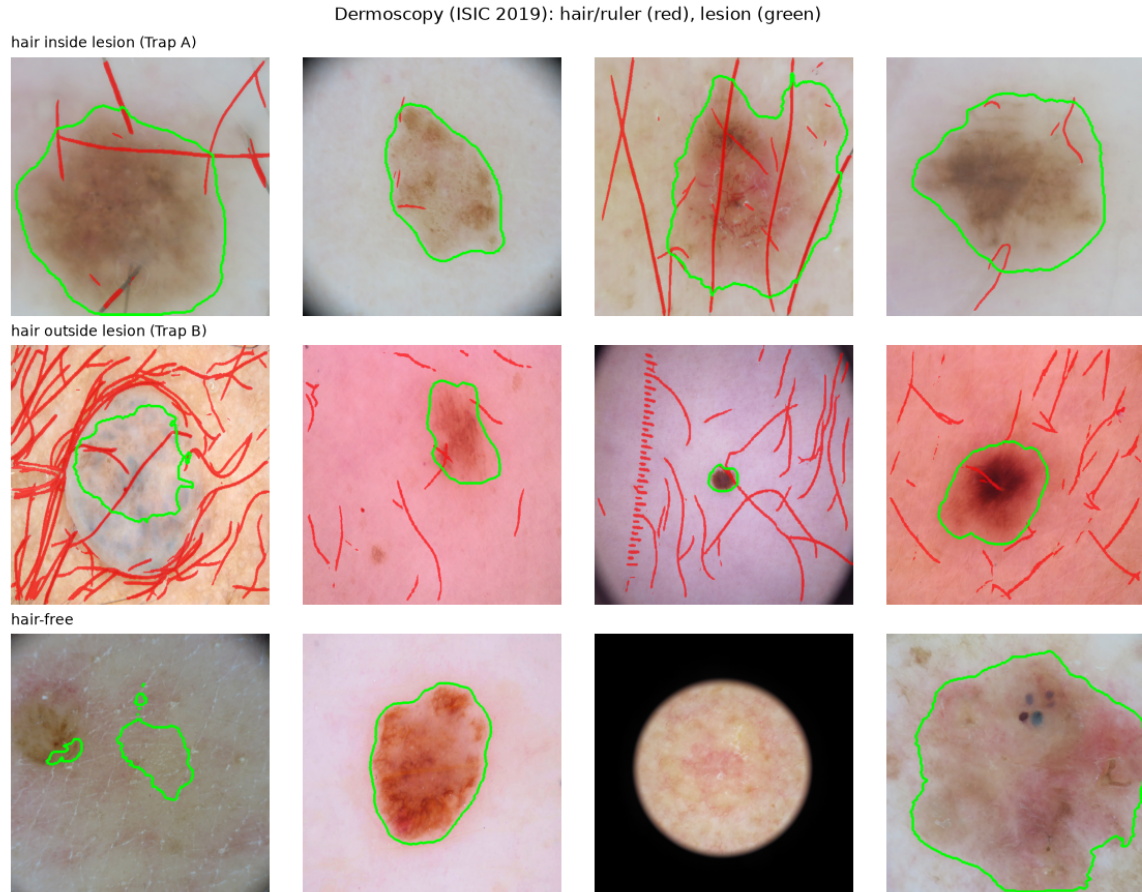


Figure 1: Dermoscopy, ISIC 2019 (25,331 images). Hair and ruler masks from Wegley et al. (2026). Note that Trap B images can still carry some hair on the lesion: the rule is that less than 10% of the hair lies inside.

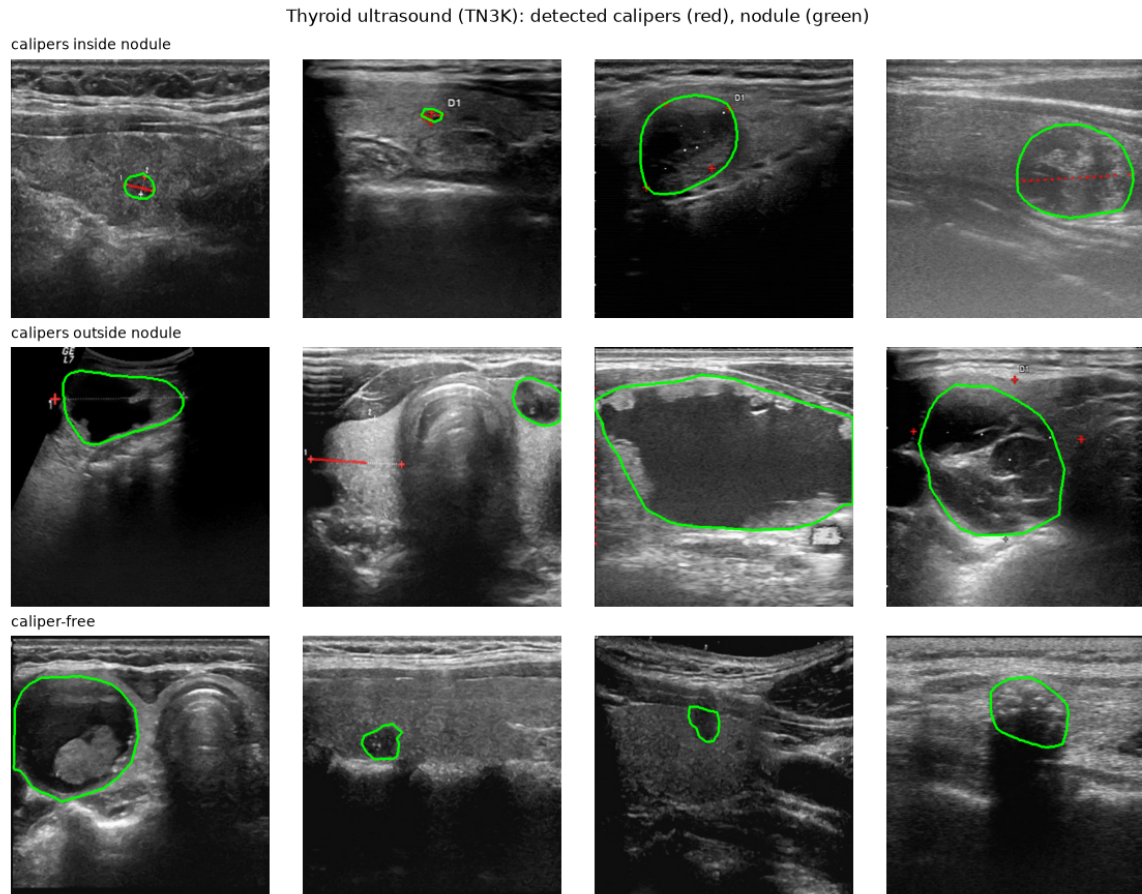
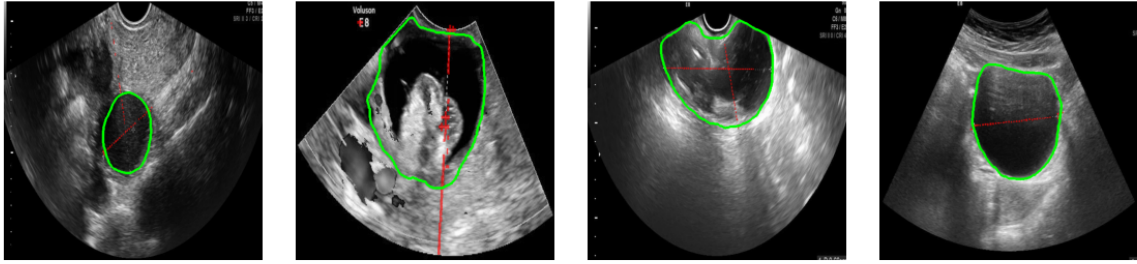


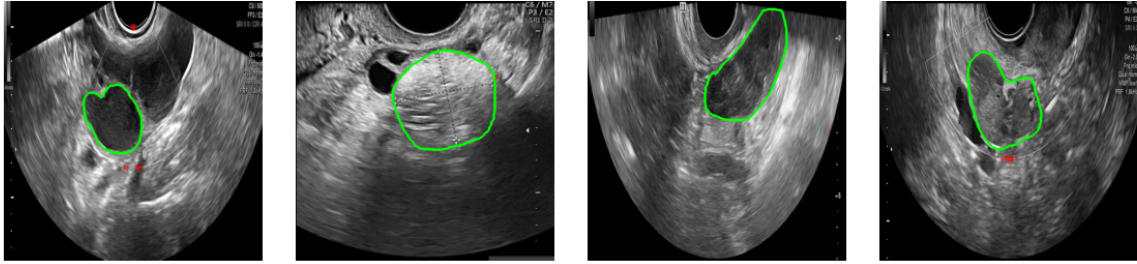
Figure 2: Thyroid ultrasound, TN3K with TNCD pathology labels (3,493 images). Calipers found by a frozen rule-based detector (95% of detections are real calipers in a 40-image audit).

Ovarian tumour ultrasound (MMOTU): calipers/overlays (red), tumour (green)

calipers inside tumour



overlays outside tumour



caliper-free

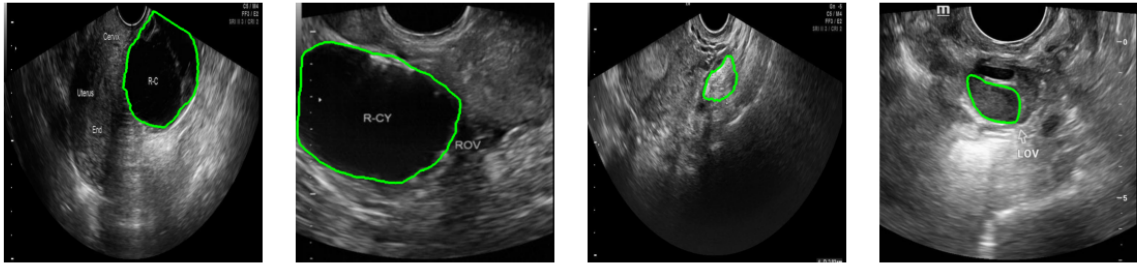


Figure 3: Ovarian tumour ultrasound, MMOTU (1,202 images; CC BY 4.0). The same detector; “outside” detections are often other on-screen overlays (zoom boxes, focus markers).

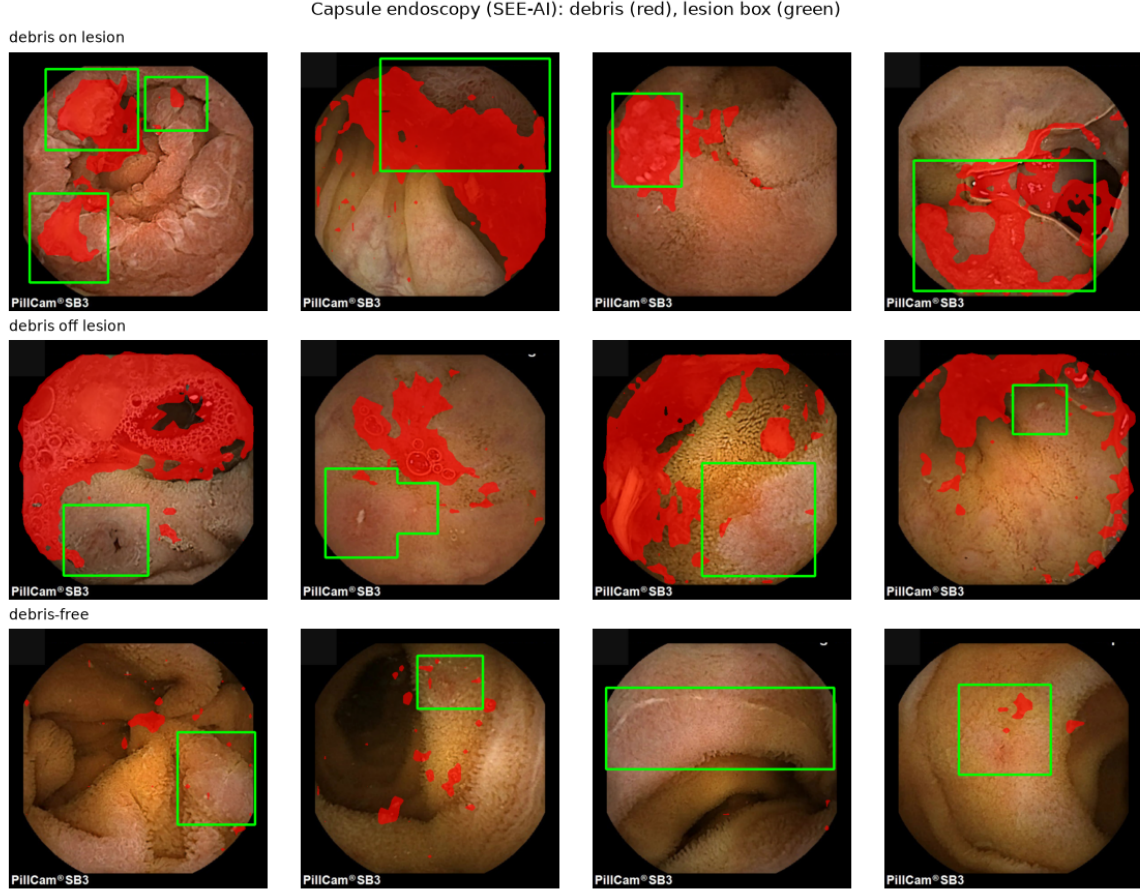


Figure 4: Capsule endoscopy, SEE-AI (5,481 frames; CC BY 4.0). Debris segmented by a patch-token probe trained on expert masks (pixel accuracy 0.92, IoU 0.62). Debris can resemble the yellow-white fibrin of erosions.

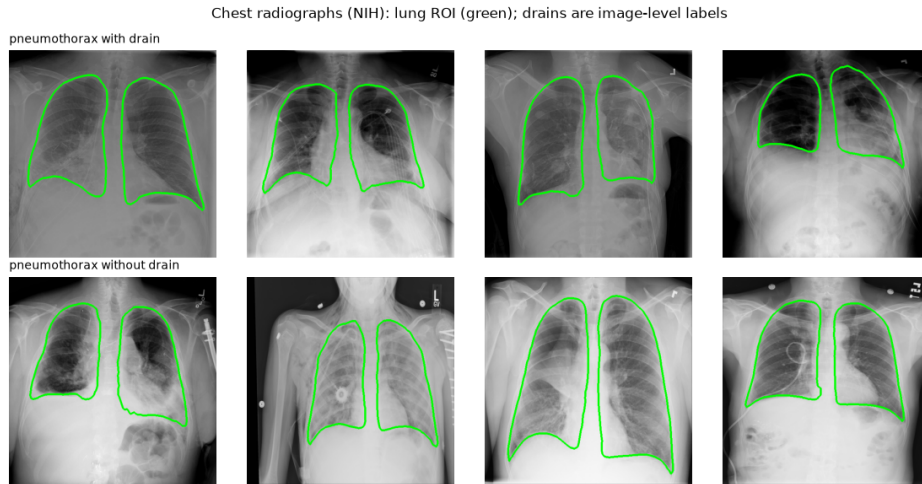


Figure 5: Chest radiographs, NIH ChestX-ray14. Drains are labelled per image (no pixel masks); they lie inside the lungs, so only the in-ROI case exists.

Controlled (synthetic) artifacts — described, not shown. To isolate location, the controlled experiments paste a drawn artifact (ruler for dermoscopy, tube for chest X-ray, dotted caliper for ultrasound, debris blob for capsule) onto real artifact-free images at 0/25/50/75/100% overlap with the ROI. U-MtE uses a separate generic overlay library (random strokes, bands, patches, blobs, dashed

lines, glyphs) that never contains a real artifact type. All images shown in this report are real, unmodified clinical images with their real artifacts.

4 Methods in brief

In plain words. We never retrain the big image models in the main analysis; we take their image descriptions and train a simple classifier on top, which lets us test many fixes cleanly. We then confirm the key results by training whole networks.

- **Backbones:** DINOv2 ViT-B/14, DermLIP (dermatology), MedSigLIP (medical), RAD-DINO (chest X-ray), ConvNeXt-Base (CNN); fine-tuning with ResNet-50 and ViT-S.
- **Mitigations compared:** ROI masking; inpainting (classical Telea and modern LaMa); group balancing; DFR; GroupDRO; JTT; LEACE; SPLINCE; prevalence calibration; SLAS (patch-token suppression); U-MtE and its variants.
- **U-MtE:** mask, insert generic overlays into masked training images, estimate the feature subspace the overlays move, erase it, train the classifier. *Protected* U-MtE first removes from that subspace the directions that separate diseases on artifact-free images.
- **Statistics:** hierarchical paired bootstrap (10,000), clusters = seeds or folds; Holm correction over the 16 headline tests; pre-registration before every experiment (`docs/PREREGISTRATION_*.md`).

5 Results

5.1 Replicating the original pilot

Succeeded 73 of 88 numbers from the thesis proposal reproduced exactly, 12 more in sign and significance, 3 differ (documented in `results/SUMMARY.md`).

5.2 The location effect

In plain words. Blue bars are supported after strict multiple-testing correction. The first panel is the main finding: in every disease, masking is worth much more when the artifact is outside the ROI.

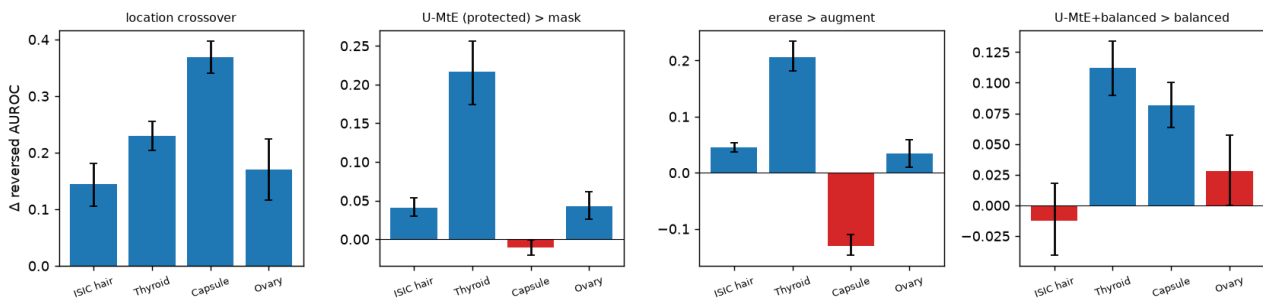


Figure 6: The 16 pre-registered headline tests (Holm-corrected; 12/16 supported).

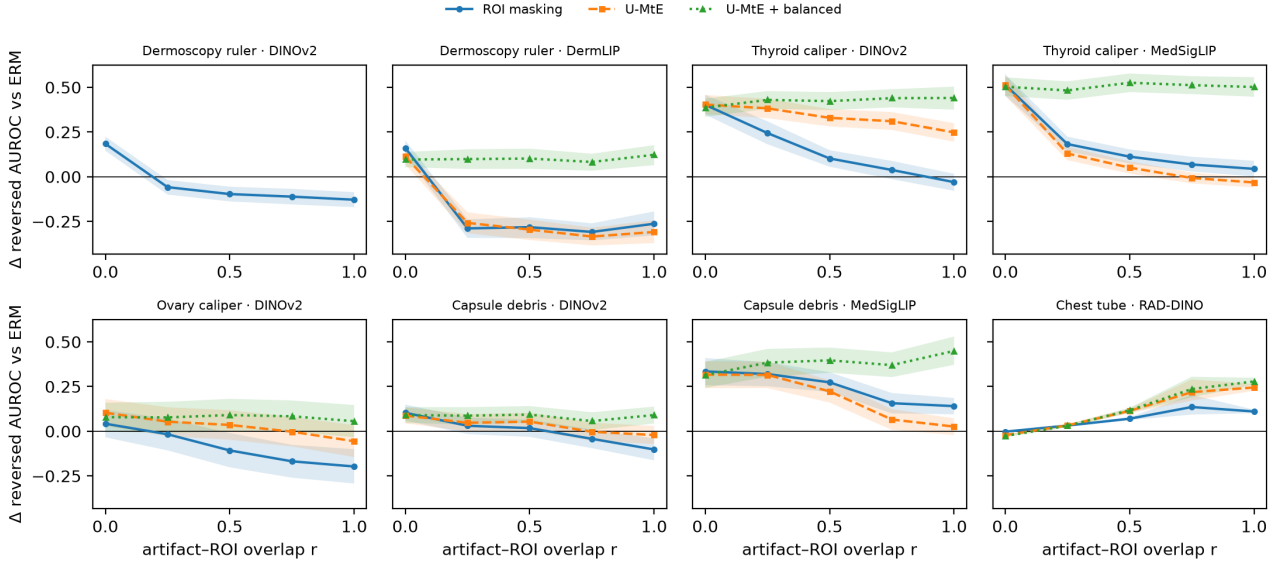


Figure 7: Controlled sweeps: effect of each method versus ERM as the artifact moves into the ROI. Masking (blue) falls; U-MtE + balancing (green) stays flat. The chest-tube panel is the boundary case in which the model ignores out-of-lung tubes altogether.

5.3 On real, unaltered data

In plain words. We trained on data exactly as the hospitals released it and tested on untouched data—including testing on a hospital the model never saw. The question: on the naturally occurring “hard” cases (where the artifact points the wrong way), does masking hurt?

Natural test	masking vs plain model (hard cases)	ours + balancing vs masking	best method (hard cases)
Thyroid, official split	−0.102 [−0.139, −0.066]	—	balancing
Melanoma, BCN held out	−0.016 [−0.027, −0.005]	+0.034	balancing $\approx$ ours+bal.
Melanoma, MSK held out	−0.056 [−0.084, −0.029]	+0.041	balancing
Melanoma, HAM held out	+0.053 [+0.037, +0.068]	+0.065	ours + balancing
Capsule, held-out videos	+0.021 [+0.000, +0.044]	+0.014	ours + balancing

Mixed Masking harms the naturally occurring hard cases in 3 of 5 tests (all three significant). Our method with balancing improves on masking for these cases in every test where it was measured, and is the best or tied-best in 3 of 5; plain balancing is best in 2 of 5.

5.4 Foundation models (no training)

In plain words. We asked the big medical AI models to classify using text prompts alone, with no training at all. Their answers are already swayed by the artifacts, and masking helps them only when the artifact is outside the lesion—the same location law, built into the models themselves.

Succeeded Masking’s effect on zero-shot hard-test accuracy is worse for in-ROI than out-of-ROI artifacts in all four tests (thyroid +0.10, capsule +0.05, ovarian +0.18, hair +0.05 location gaps, all significant).

Failed Text-prompted erasure (describe the artifact in words, erase that direction): a near no-op everywhere ($\leq +0.008$). Our image-side method with the same model gains +0.20 (thyroid) and +0.22 (ovarian).

5.5 Training the whole network

Mixed Capsule (ResNet-50): masking harms in-ROI (−0.28), crossover +0.58. Thyroid (ViT-S): crossover +0.17 [+0.07, +0.27]. End-to-end U-MtE: +0.14 over masking with ResNet-50 on thyroid,

no gain with ViT-S or on capsule.

5.6 Why: a simple theory

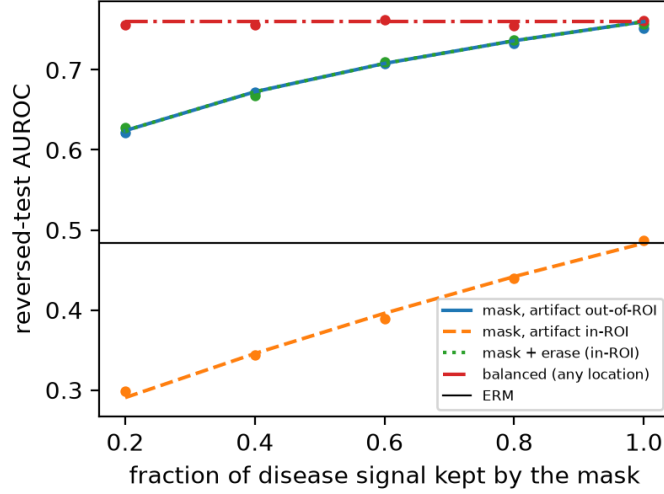


Figure 8: $\text{AUROC}_{\text{rev}} = \Phi((S - \tilde{A})/\sqrt{2(S + \tilde{A})})$ with disease signal S and effective artifact strength $\tilde{A}$ (lines) versus simulation (points).

In plain words. Masking an in-ROI artifact keeps the artifact and throws away other disease evidence, so the model leans on the artifact even more. Erasing works unless the artifact looks like the disease. Re-balancing works wherever the artifact sits.

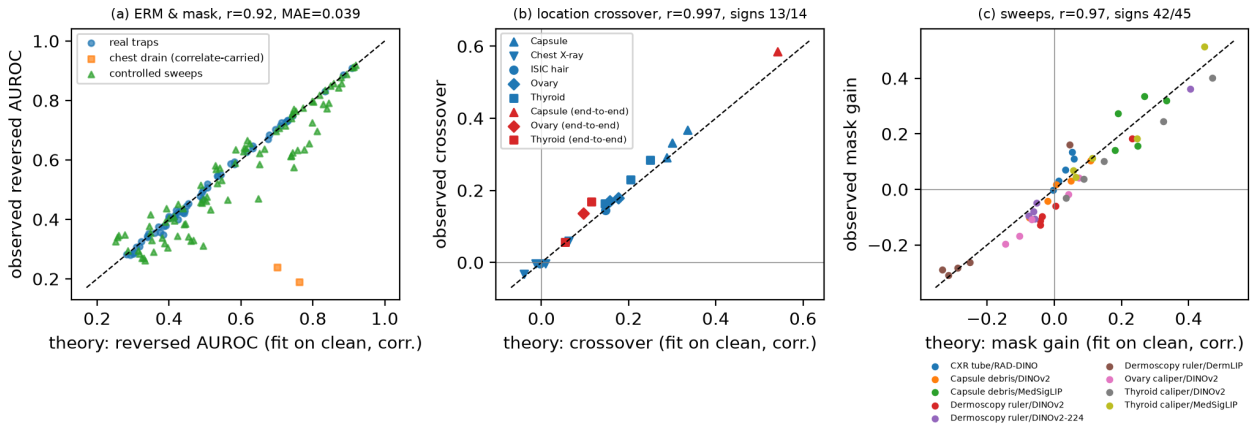


Figure 9: Pre-registered test on real models: the formula is fitted to each model’s clean and correlated results, and its hard-test (reversed) results are compared with the real ones, which it never saw.

In plain words. **Succeeded** The formula matches real models closely (left: average error 0.039; middle: the location gap, correlation 0.997). **Failed** It misses the chest-drain models, where the shortcut lives in things that come with the drain (a treated lung), which the formula cannot see. Because one of 14 location-gap signs (a value of -0.002) disagreed, our pre-registered rule calls it a “qualitative” account.

Table 1: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by `scripts/make_main_table.py`.

Method	Needs	Hair/DINOv2	Hair/DerMLIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNeXt	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNeXt	Drain/RAD-DINO	Drain/DINOv2
ERM	—	0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)	—	0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)	—	0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MtE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MtE, protected (ours)	ROI + Y of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.66)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)	—	0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)	—	0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)	—	0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)
U-MtE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

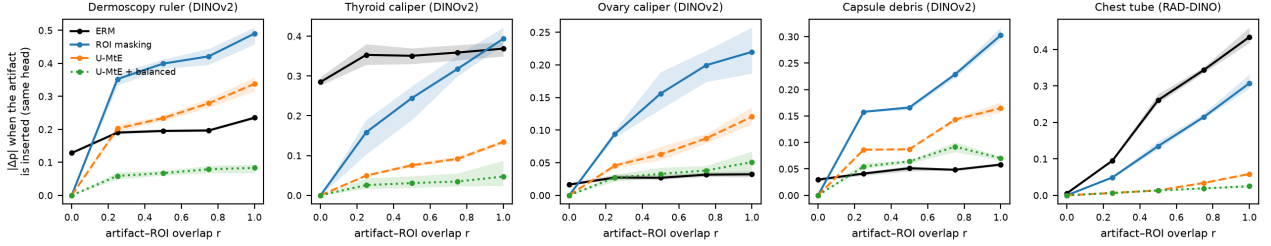


Figure 10: Why masking fails in the ROI: how much the prediction changes when the artifact is added, versus how deep inside the ROI it sits. Masking (blue) makes the model rely on an in-ROI artifact more; U-MtE (orange) and U-MtE with balancing (green) remove that reliance.

5.7 Remedies compared

In plain words. Each column is a dataset/model; each row a method. Numbers are hard-test accuracy (in brackets: the fairer summary that also penalises “flipping”). Bold = best in the column.

- **Succeeded** Erase beats augment with the same overlays: thyroid +0.21, hair +0.05, ovary +0.04; **Failed** capsule −0.13.
- **Succeeded** SPLINCE (closest published erasure, NeurIPS 2025) removes much disease signal and flips the shortcut on capsule; U-MtE + balancing is more robust in 3 of 4 cohorts (not on chest drains, where DFR is best).
- **Mixed** LaMa inpainting with artifact masks is strong; protected U-MtE matches it on thyroid without masks.
- **Mixed** JTT (label-free): U-MtE wins on thyroid (+0.20) and ovary (+0.14); JTT wins on hair.
- **Mixed** Selector: beats masking in all 5 cohorts, within 0.02 of the best candidate, beats DFR in 3/5.

5.8 Robustness

Succeeded Visual audit (40 images per cell) and re-runs with large-marker and strict-location definitions: the location effect and the protected-U-MtE gain hold in every variant (`docs/SENSITIVITY_ARTIFACT_LABELS.md`).

Succeeded Stricter patient-leakage control. Thyroid, ovary and capsule datasets have no patient IDs. We also grouped images whose foundation-model embeddings are near-identical (same patient or session) and re-ran everything: 11 of 12 main results are unchanged; **Failed** only the small ovary “erase beats augment” advantage loses significance.

Succeeded Model size. Small, base and large DINOv2 models (21 M to 304 M parameters): the location effect holds in all 9 dataset/size pairs, and U-MtE beats masking on thyroid and ovary at every size. Larger models cheat *more* on in-lesion calipers, so bigger models do not fix the problem.

Succeeded Erasure rank. U-MtE’s thyroid gain is +0.18 to +0.22 whenever it erases 16 or more directions; the protected version never hurts capsule by more than 0.01.

Failed Fine-tune, then erase. Applying U-MtE after fully fine-tuning a ResNet-50 does not help (+0.004): U-MtE is a tool for frozen foundation-model features.

Succeeded Erase while fine-tuning. Doing the erasure *during* fine-tuning (on the network’s own features, plus making features and predictions ignore overlays) works for both a CNN and a transformer on thyroid: ResNet-50 +0.157, ViT-S +0.143 over masking, with no artifact annotation. **Mixed** Ovary: +0.094 (positive, not significant; small dataset). **Failed** Capsule: fails (debris looks like the disease), as expected.

Mixed Clinical metrics. On the untouched thyroid test set, masking halves sensitivity at a fixed operating point ($0.73 \rightarrow 0.35$) while raising specificity ($0.62 \rightarrow 0.89$); U-MtE recovers part of it and has the best AUPRC and Brier score. On other hospitals and capsule, masking improves every metric.

6 What did not work (and why it is still useful)

- **Failed** “Masking always harms in-ROI” is not universal: in real thyroid/ovary/capsule it helps a little (removing clutter); in chest X-rays the model ignores out-of-lung tubes. The universal claim is the location *gap*.
- **Failed** U-MtE alone on capsule debris (resembles erosion fibrin) and with MedSigLIP in controlled sweeps.
- **Failed** SLAS (patch suppression): localises hair from 5 annotations but gives small classification gains.
- **Failed** Pseudo-group balancing without labels: helps thyroid only.
- **Failed** Datasets dropped: CANDID-PTX (no access), breast ultrasound (unreliable caliper labels), colonoscopy (no reflection-free images), ink (too few), cardiomegaly lines (too few), pathology pen marks (unreliable masks).

7 Reproduce and reuse

make data data_extra, then one target per experiment (README.md); every number is written by a script. Tools: `python -m wtss.umte fit/embed` and `python -m wtss.slas fit/embed` for any backbone.

Data attribution

ISIC Archive (CC BY-NC); HAM10000 (Tschandl et al.); Wegley et al. 2026 artifact masks; TN3K (Gong et al.) with TNCD labels; MMOTU (Zhao et al., CC BY 4.0); SEE-AI (CC BY 4.0) and figshare 27645021; NIH ChestX-ray14; NEATX; RANZCR-CLiP. Images appear as thumbnails for scientific illustration.